

Default genomic exon-span CTMC on a supplied tree

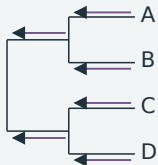
One physical exon-span geometry with declared DNA-material states

A Tree and tip evidence



Physical exon-span compatibility
Root, topology and branch lengths
are supplied model conditions

B Conditional pruning and family likelihood



Prune within one origin scenario s

$$L_u(s) = \sum_i \pi_i(s) L_{\text{root}}(i | s)$$

Scenario likelihoods are weighted by their prior

$$L_u = \sum_s p(s) L_u(s)$$

Family composite log likelihood

$$\ell_{\text{family}} = \sum_u \log L_u$$

$$P_{s,b}(t) = \exp(Q_{s,b}t)$$

Node marginals integrate origin scenarios

C Branch endpoint summaries



Branch P: geometry differs

Joint modal endpoint pair

D Per-family rate fit

One nonnegative μ per family

Shared across edit kinds
and linked local units

Linked units: composite likelihood

E Declared conditions

Supplied tree + branch lengths
Finite exon-boundary catalogue
Root and origin assumptions

Inference conditional on
these declared choices

$$q_{ij} = \mu \sum_{e:i \rightarrow j} w_e \quad (i \neq j)$$

$$q_{ii} = -\sum_{j \neq i} q_{ij}$$

μ is shared across the edit graph; weights divide opportunities.

Equal-rate graph baseline $\neq$ biological rate equality; endpoints do not specify paths.